AI-Enabled Spatial Augmented Reality for Human-Centred Manual Assembly in Industry 5.0, A PRISMA Systematic Review of Sensing, Adaptation, and TRL

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Appendix A. Database Search Strategy

Search Strategy Table can be found at https://github.com/ISILablsbu/prisma26$_NH/blob/main/searchStrategy.$

Appendix B. Appendix title 2

Table B.1: Extended Overview of SAR Studies with Adaptivity and Operator-Centric Factors (Part 1/4).

Author (Year)	Country	Domain	Study Type	SAR Focus	AI	Cognitive Monitoring	Adaptivity	Op 5.0	NASA-TLX	Maturity	Gesture	Occlusion	Product Type
Büttner et al. (2015) [1]	Germany	Generic / unspecified	Prototype	Guidance / Instructions	Rule-based	None / Vision-only	Static	No	NR	Lab Prototype	No - Foot Pedal control	NR	LEGO sets
Funk et al. (2015) [2]	Germany	Generic / unspecified	Controlled Lab	Accessibility / Inclusion	Rule-based	None / Vision-only	Process-adaptive	Aligned (Implicit)	NR	Lab Prototype	No - Pick task recog.	NR	LEGO Duplo
Funk et al. (2015) [3]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Spatial-adaptive	No	20.25	Lab Prototype	No - Pick task recog.	NR	LEGO bricks
Funk et al. (2015) [4]	Germany	Generic / unspecified	Field Study	Guidance / Training	Rule-based	Subjective (self-report)	Process-adaptive	Aligned (Implicit)	44	Industrial Pilot	No - Pick task recog.	NR	Clamp
Funk et al. (2015) [5]	Germany	Automotive	Field Deployment	Quality / Monitoring	Rule-based	None / Vision-only	Process-adaptive	No	NR	Industrial Pilot	No - Task recog.	NR	Engine starter
Funk et al. (2015) [6]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Static	No	33.16 (workload)	Lab Prototype	No	NR	LEGO Duplo / Wood
Büttner et al. (2016) [7]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	None / Vision-only	Static	No	NR	Lab Prototype	No	NR	LEGO sets
Fiorentino et al. (2016) [8]	Italy	Automotive	Field Study	Guidance / Instructions	Rule-based	None / Vision-only	Static	No	NR	Field Study	Yes	Depth Separation	BMW Z4
Funk et al. (2016) [9]	Germany	Generic / unspecified	Field Studies	Guidance / Training	Rule-based	None / Vision-only	Process-adaptive	No	NR	Industrial Pilot	No - Activity recog.	NR	Auto parts
Funk et al. (2016) [10]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Spatial-adaptive	No	25.37	Lab Prototype	No -Head Tracking	Hardware Avoidance	LEGO bricks
Funk et al. (2016) [11]	Germany	Generic / unspecified	Controlled Lab	Quality / Monitoring	Rule-based	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No - Pick task recog.	NR	LEGO Duplo
Funk et al. (2016) [12]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Static	No	28.13	Lab Prototype	No - Task recog.	NR	LEGO Duplo
Kosch et al. (2016) [13]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	None / Vision-only	Process-adaptive	Aligned (Implicit)	NR	Lab Prototype	No - Pick task recog.	NR	LEGO Duplo
Mura et al. (2016) [14]	Italy	Generic / unspecified	Lab Prototype	Quality / Monitoring	Rule-based	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No - weight sensor	NR	Assembly set
Obinata et al. (2016) [15]	Japan	Generic / unspecified	Tech Validation	Guidance / Instructions	Rule-based	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	No	Hand removal algo.	PC / Box
Sand et al. (2016) [16]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	No - Foot Pedal control	NR	LEGO sets
Xu et al. (2017) [17]	China	Aerospace	Lab Exp	Guidance / Instructions	Machine Learning	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	No	NR	Aircraft Model
Bosch et al. (2017) [18]	Netherlands	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Static	No	36.8 (workload)	Lab Prototype	No - Task recog.	NR	Abstract Task
Büttner et al. (2017) [19]	Germany	Automotive	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Process-adaptive	No	31.5, 23.5	Lab Prototype	Yes	NR	Belt Tensioner
Funk et al. (2017) [20]	Germany	Automotive	Field Study	Guidance / Instructions	Rule-based	Subjective (self-report)	Process-adaptive	No	74.34	Industrial Pilot	No - Pick task recog.	NR	Engine Starters
Kosch et al. (2017) [21]	Germany	Automotive	Reflection	Guidance / Instructions	Rule-based	None / Vision-only	Process-adaptive	Aligned (Implicit)	NR	Pilot	No	NR	Engine starter
Blattgerste et al. (2018) [22]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Static	No	29.6	Lab Prototype	No - Clicker	Occlusion aware	LEGO Duplo
Funk et al. (2018) [23]	Germany	Automotive	Field Study	Training	Rule-based	Subjective (self-report)	Process-adaptive	No	15.62	Industrial Pilot	Yes	NR	Engine starter
Kosch et al. (2018) [24]	Germany	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Physiological (EEG/EDA)	Static	No	24.17	Lab Prototype	No - Pick task recog.	NR	LEGO Duplo
Mengoni et al. (2018) [25]	Italy	Consumer Electronics	Controlled Lab	Human factors / Safety	Rule-based	Subjective (self-report)	Process-adaptive	Aligned (Implicit)	26.17	Lab Prototype	No	NR	Smartphone

Table B.2: Extended Overview of SAR Studies with Adaptivity and Operator-Centric Factors (Part 2/4).

Author (Year)	Country	Domain	Study Type	SAR Focus	AI	Cognitive Monitoring	Adaptivity	Op 5.0	NASA-TLX	Maturity	Gesture	Occlusion	Product Type
Simoes et al. (2018) [26]	Spain	Electronics	Tech Validation	Guidance / Instructions	Rule-based	None / Vision-only	Spatial-adaptive	No	NR	Proof Concept	Yes	NR	Electrical Cabinet
Uva et al. (2018) [27]	Italy	Automotive	Controlled Lab	Guidance / Instructions	Rule-based	None / Vision-only	Static	No	NR	Lab Prototype	No - Joystick	NR	Honda Engine
Alves et al. (2019) [28]	Portugal	Generic / unspecified	Controlled Lab	Quality / Monitoring	Rule-based	Subjective (self-report)	Process-adaptive	No	20	Lab Prototype	No -Wizard of oz	NR	Puzzle
Costa et al. (2019) [29]	Portugal	Automotive	Tech Validation	Training	Rule-based	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	No	NR	Starter Motor
Faccio et al. (2019) [30]	Italy	Automotive	Controlled Lab	Quality / Monitoring	Rule-based	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No - Activity recog.	NR	Water pump
Haslwanter et al. (2019) [31]	Austria	Generic / unspecified	Controlled Lab	Other	Machine Learning	None / Vision-only	Content-adaptive	Op 4.0	NR	Lab Prototype	No - Button glove	NR	LEGO Duplo
Kosch et al. (2019) [32]	Germany	Generic / unspecified	Controlled Lab	Human factors / Safety	Rule-based	Physiological (EEG/EDA)	Static	No	NR	Lab Prototype	No - Task recog.	NR	LEGO Duplo
Wang et al. (2019) [33]	China	Automotive	Controlled Lab	Remote collaboration	Rule-based	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	Yes	NR	Water pump
Wilschut et al. (2019) [34]	Netherlands	Industrial equipment	Controlled Lab	Training	Rule-based	Subjective (self-report)	Static	No	28.1	Lab Prototype	No - Pick Tracking	NR	Pneumatic
Bosch et al. (2020) [35]	Netherlands	Generic / unspecified	Field Study	Guidance / Training	Rule-based	Subjective (self-report)	Static	Op 4.0	31	Industrial Pilot	No - Bin Picking	NR	Tooling
Büttner et al. (2020) [36]	Germany	Industrial equipment	Controlled Lab	Training	Rule-based	None / Vision-only	Static	No	NR	Lab Prototype	No - Wizard of oz	NR	Pneumatic
Heinz et al. (2020) [37]	Germany	Genric/unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Behavioural	Static	No	NR	Lab Prototype	Yes	NR	Toy Gearbox
Horejsi et al. (2020) [38]	Czech Republic	Generic / unspecified	Controlled Lab	Training	Rule-based	None / Vision-only	Static	No	NR	Industrial Pilot	No - Buttons	NR	Sink trap
Mark et al. (2020) [39]	Italy	Industrial equipment	Controlled Lab	Training	Rule-based	None / Vision-only	Static	No	NR	Lab Exp	No -Buttons	NR	Pneumatic Cylinder
Ojer et al. (2020) [40]	Spain	Electronics	Lab Test	Guidance / Instructions	Machine Learning	None / Vision-only	Spatial-adaptive	Op 4.0	90 (SUS)	Lab Prototype	No	NR	PCB
Rupprecht et al. (2020) [41]	Austria	Aerospace	Controlled Lab	Guidance / Instructions	Rule-based	None / Vision-only	Spatial-adaptive	Op 4.0	NR	Lab Prototype	No - Task recog.	NR	Aircraft Components
Serras et al. (2020) [42]	Spain	Generic / unspecified	User Study	Guidance / Instructions	Rule-based	None / Vision-only	Content-adaptive	Op 4.0	NR	Industrial Pilot	No -Speech/Voice	NR	Gripper / Panel
Smith et al. (2020) [43]	UK	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Static	No	1.845	Lab Exp	No -Touch/Voice	NR	LEGO bricks
Vanneste et al. (2020) [44]	Belgium	Generic / unspecified	Controlled Lab	Human factors / Safety	Rule-based	Subjective (self-report)	Static	No	NR	Lab Exp	No - Buttons	NR	Light fixture
Wang et al. (2020) [45]	China	Generic / unspecified	Controlled Lab	Remote collaboration	Rule-based	Behavioural	Spatial-adaptive	No	NR(51)	Lab Prototype	No - Head / Eye Gaze	NR	LEGO Duplo
Wang et al. (2020) [46]	China	Generic/unspecified	Controlled Lab	Guidance	Rule-based	Subjective (self-report)	Static	No	NR	Lab Prototype	No	NR	Square Plugs/Tubes
Chen et al. (2021) [47]	China	Generic / unspecified	Tech Validation	Quality / Monitoring	Machine Learning	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	Yes	Pixel Classification	Gear retarder
Froschauer et al. (2021) [48]	Austria	Generic / unspecified	Tech Validation	Guidance / Instructions	Deep Learning	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No	NR	LEGO / Wood
Heinz et al. (2021) [49]	Germany	Electronics	Field Study	Guidance / Instructions	Rule-based	None / Vision-only	Process-adaptive	Aligned (Implicit)	NR	Industrial Pilot	No - Pick task recog.	NR	PCB
Pimminger et al. (2021) [50]	Austria	Electronics	Field Study	Guidance / Instructions	Deep Learning	None / Vision-only	Spatial-adaptive	No	NR	Industrial Pilot	No - Activity recog.	NR	Ringbandkern
Rupprecht et al. (2021) [51]	Austria	Aerospace	Controlled Lab	Guidance / Instructions	Deep Learning	Subjective (self-report)	Spatial-adaptive	No	32.2	Lab Prototype	Yes	NR	Fan cowl

Table B.3: Extended Overview of SAR Studies with Adaptivity and Operator-Centric Factors (Part 3/4).

Author (Year)	Country	Domain	Study Type	SAR Focus	AI	Cognitive Monitoring	Adaptivity	Op 5.0	NASA-TLX	Maturity	Gesture	Occlusion	Product Type
Schmitt et al. (2021) [52]	Germany	Generic / unspecified	Tech Validation	Guidance / Instructions	Rule-based	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No - Task/Head Tracking	NR	Toy Truck
Schmitt et al. (2021) [53]	Germany	Generic/unspecified	Tech Demo	Guidance	Rule-based	None / Vision-only	Process-adaptive	No	NR	Demonstration	No	NR	Toy Truck
Sehr et al. (2021) [54]	Germany	Generic / unspecified	Controlled Lab	Adaptive Guidance	Rule-based	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No - Task recog.	NR	Block Structure
Simoes et al. (2021) [55]	Spain	Electronics	Field Study	Training	Rule-based	None / Vision-only	Content-adaptive	Aligned (Implicit)	NR	Industrial Pilot	No - Task recog.	NR	Cabinet
Simonetto et al. (2021) [56]	Norway	Generic / unspecified	Controlled Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Process-adaptive	No	86.7%	Lab Prototype	No	NR	LEGO model
Wang et al. (2021) [57]	China	Automotive	Controlled Lab	Remote collaboration	Rule-based	Subjective (self-report)	Process-adaptive	No	47	Lab Prototype	No -Head/Eye gaze	NR	Water pump
Wang et al. (2021) [58]	China	Generic / unspecified	Controlled Lab	Remote collaboration	Rule-based	Behavioural	Other adaptive	No	NR	Lab Prototype	Yes	Active Apparent	Gear reducer
Fang et al. (2022) [59]	China	Generic / unspecified	Controlled Lab	Guidance / Instructions	Deep Learning	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	Yes	DNN Mask	Reducer
Heinz-Jakobs et al. (2022) [60]	Germany	Consumer goods	Field Study	Accessibility / Inclusion	Rule-based	None / Vision-only	Process-adaptive	Aligned (Implicit)	NR	Industrial Pilot	No - Pick task recog.	NR	Bottle opener
Hercog et al. (2022) [61]	Slovenia	Generic / unspecified	Controlled Lab	Guidance / Instructions	Deep Learning	None / Vision-only	Static	No	NR	Prototype	No - PtL/CV	NR	Puzzle
Moreira et al. (2022) [62]	UK	Automotive	Controlled Experiment	Training	Rule-based	None / Vision-only	Process-adaptive	Yes	NR	Industrial Pilot	No	NR	EV Battery
Rupprecht et al. (2022) [63]	Austria	Generic / unspecified	Lab Study	Guidance / Instructions	Deep Learning	Subjective (self-report)	Spatial-adaptive	No	33.2	Industrial Pilot	Yes	NR	Toy Bricks
Wang et al. (2022) [64]	China	Automotive	Controlled Lab	Remote collaboration	Rule-based	Behavioural	Spatial-adaptive	No	NR	Lab Prototype	Yes	NR	Water Pump
Zhang et al. (2022) [65]	China	Aerospace	Controlled Lab	Guidance / Instructions	Rule-based	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	Yes	Thresholding	Aero Blade
Zigart et al. (2022) [66]	Austria	Electronics	Field/Lab	Guidance / Instructions	Rule-based	Subjective (self-report)	Static	No	19.38	Industrial Pilot	No - Buttons	NR	Control Panel
Moghaddam et al. (2023) [67]	Belgium	Generic / unspecified	Tech Validation	Guidance / Instructions	Deep Learning	Behavioural	Other adaptive	Yes	NR	Validation	No	NR	Manual Steps
Chojecki et al. (2023) [68]	Germany	Automotive	Controlled Lab	Guidance / Instructions	Deep Learning	Subjective (self-report)	Static	No	7.254 (Frus.)	Lab Prototype	Yes (Multimodal)	Segmentation	Car subframe
David et al. (2023) [69]	Finland	Automotive	Controlled Lab	Guidance / Intent	Rule-based	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	No	NR	Diesel engine
Gerhard et al. (2023) [70]	Germany	Generic / unspecified	Tech Validation	Guidance / Instructions	Rule-based	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	Yes	NR	Robot chassis
Khafaga et al. (2023) [71]	UK	Automotive	Controlled Lab	Guidance	Rule-based	Subjective (self-report)	Static	Yes	NR	Lab Prototype	No	NR	E-motor
Pasaribu et al. (2023) [72]	Indonesia	Generic/unspecified	Controlled Lab	Guidance	Rule-based	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No - RFID tags	NR	LEGO bricks
Riedel et al. (2023) [73]	Germany	Consumer electronics	Controlled Lab	Guidance / Instructions	Deep Learning	Subjective (self-report)	Process-adaptive	No	7.04	Lab Prototype	No	NR	Tube Lamp
Stübl et al. (2023) [74]	Austria	Consumer goods (other)	Field Study	Quality / Monitoring	Deep Learning	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No	NR	Furniture
Wang et al. (2023) [75]	China	Generic/unspecified	Controlled Lab	Remote collaboration	Rule-based	Subjective (self-report)	Static	No	P(0.594)	Lab Prototype	Yes	NR	3D parts
Wedral et al. (2023) [76]	Austria	Generic/unspecified	Controlled Lab	Safety / Ergonomics	Deep Learning	Subjective (self-report)	Spatial-adaptive	Yes	21.488	Factory Lab	No	NR	Clamps
Haas et al. (2024) [77]	Germany	Industrial equipment	Concept	Ergonomics	Machine Learning	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	No	NR	Fuel Cell

Table B.4: Extended Overview of SAR Studies with Adaptivity and Operator-Centric Factors (Part 4/4).

Author (Year)	Country	Domain	Study Type	SAR Focus	AI	Cognitive Monitoring	Adaptivity	Op 5.0	NASA-TLX	Maturity	Gesture	Occlusion	Product Type
Liang et al. (2024) [78]	New Zealand	Industrial equipment	Tech Validation	Guidance	Deep Learning	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	No - Activity recog.	NR	Hydraulic valve
Lucchese et al. (2024) [79]	Norway	Generic/unspecified	Controlled Lab	Guidance	Rule-based	None / Vision-only	Process-adaptive	Yes	NR	Lab Exp	No - Task recog.	NR	LEGO models
Stoll et al. (2024) [80]	Germany	Consumer electronics	Tech Validation	Guidance	Rule-based	None / Vision-only	Multi-adaptive	Yes	NR	Lab Prototype	Yes	NR	Smartphone
Wachter et al. (2024) [81]	Germany	Generic/unspecified	Lab Exp	Guidance	Deep Learning	None / Vision-only	Process-adaptive	No	NR	Lab Prototype	Yes	Segmentation Fusion	Picking
Wang et al. (2024) [82]	China	Aerospace	Tech Validation	Guidance	Rule-based	None / Vision-only	Spatial-adaptive	No	NR	Lab Prototype	No	NR	Aircraft Cabin
Hasan et al. (2025) [83]	UK	Generic/unspecified	Controlled Lab	Interactive Guidance	Deep Learning	Subjective (self-report)	Process-adaptive	Yes	2.10/5	Lab Prototype	Yes	NR	LEGO Duplo
Kelm et al. (2025) [84]	Germany	Industrial equipment	Tech Validation	Instructions	LLM (GPT-4)	None / Vision-only	Content-adaptive	No	NR	Software	No	NR	Fuel Cell
Lucchese et al. (2025) [85]	Norway	Generic/unspecified	Controlled Lab	Guidance	Rule-based	Subjective (self-report)	Static	Yes	43.66	Lab Prototype	No - Task recog.	NR	LEGO Bricks
Pang et al. (2025) [86]	China	Industrial equipment	Tech Validation	Flexible Info	Rule-based	None / Vision-only	Spatial-adaptive	Yes	NR	Simulation	No	Layout optimisation	Thin-walled
Seliger et al. (2025) [87]	Germany	Industrial equipment	Tech Validation	Guidance	Deep Learning	None / Vision-only	Spatial-adaptive	No	NR	Proof Concept	No	NR	Ceramics
Simeone et al. (2025) [88]	Italy	Industrial equipment	Controlled Lab	Guidance	GenAI / Hybrid	None / Vision-only	Other adaptive	No	NR	Lab Setup	No	NR	Centrifugal Pump
Zhang et al. (2025) [89]	China	Aerospace	Controlled Lab	Guidance	Rule-based	Subjective (self-report)	Process-adaptive	No	23.16	Lab Prototype	No	NR	Aircraft Panel